

ColonAI — Technical Handover

A complete record of every decision, file, metric, known bug and improvement idea — for resuming this work in any future session, with or without me.

Project status	Live demo running. v2 model deployed.
Last training run	epoch-4 deploy-grade retrain — val acc 0.865, cross-vendor IoU mean 0.268, Pentax 0.16
Active checkpoint	outputs/unified_multimodal_v2/checkpoints/best_model.pth
Calibration	T = 0.45 (ECE 0.062 on val)
Segmentation decoder	outputs/unified_multimodal_v2/seg_head.pth — IoU 0.61 cross-vendor
OOD head	outputs/unified_multimodal_v2/ood_head.pth — val F1 = 1.0
GitHub	github.com/Yuvraj235/Agentic_Multimodal_Colon_Cancer_AI (main: cb797a2)
Live Space	huggingface.co/spaces/Yuvraj2319/colonai
Model repo	huggingface.co/Yuvraj2319/colonai-v2
Auth token	stored in ~/.cache/huggingface/token (Write scope)

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1. Snapshot of the system today

Two-minute orientation if you're picking this up cold.

What works

- ✓ 5-class pathology classifier (polyps / uc-mild / uc-mod-sev / barretts-esoph / therapeutic). Val acc 86.5%, calibrated ECE 0.062.
- ✓ GradCAM++ heat-map that lands in the polyp region (not on scope artefacts). Mean cross-vendor IoU 0.27, Pentax 0.16 (was 0.07 in v1).
- ✓ Polyp segmentation decoder — IoU 0.61 across all 5 external datasets, Pentax = Olympus.
- ✓ Six-agent orchestrator (image / text / tabular / fusion / xai / recommendation) wired into both Streamlit and a REST API.
- ✓ Patient-safety policy: rejects non-endoscopy uploads, abstains on low-confidence, shows otherwise. Audit log on every prediction.
- ✓ Live-video pipeline at 21 FPS on Apple Silicon with 3-frame debounce.
- ✓ Live demo on Hugging Face Spaces (Docker SDK, free CPU tier).
- ✓ Hospital-grade REST API with X-API-Key auth, CORS allow-list, sanitised errors.

What doesn't work yet

- ✗ **UC severity grading.** Moderate-severe recall 0.15. Class-balanced sampler made the model over-call "mild" UC. Needs more uc-mod-sev training data or a hierarchical loss.
- ✗ **Per-polyp detection on Pentax.** Sensitivity@IoU0.5 is 0.38 on ETIS-Larib vs 0.75 on Olympus. Segmentation is at parity but proper detection isn't.
- ✗ **Invasive cancer / Stage III-IV.** Completely out-of-distribution. The model was trained on screening-stage data only. The image-stats safety net catches these but the system cannot classify them.
- ✗ **Confidence on hospital-real data.** ECE was measured on the public-data val split. Real-world distribution will need per-site temperature re-calibration.
- ✗ **Free-tier HF Space cold-start.** First request after idle takes ~30s while the container wakes up. Not an issue if traffic is steady.

What's deployed where

Surface	Where	Notes
Streamlit web app	huggingface.co/spaces/Yuvraj2319/colonai	Docker SDK, CPU basic, free. Checkpoint embedded in repo via G
Model checkpoint	huggingface.co/Yuvraj2319/colonai-v2	Separate model repo. 572 MB. Public.
Source code	github.com/Yuvraj235/Agentic_Multimodal_Colon_Cancer_AI	Main repo. All commits include co-author trailer.
REST API (when run locally)	scripts/serve_api.py	Defaults to 127.0.0.1:8081. Auth via COLONAI_API_KEY env var.
Local Streamlit (dev)	streamlit run app.py	127.0.0.1:8501. .streamlit/config.toml caps upload at 10 MB.

2. Architecture deep-dive

How the model, agents, and safety stack fit together.

2.1 Model — UnifiedMultiModalTransformer

Defined in `src/models/unified_transformer.py`. Multi-modal classifier with three encoders feeding a fusion transformer.

Component	Implementation
Image encoder	DualBackboneImageEncoder: ResNet50 (timm) + EfficientNet-B0. Both produce 7×7 spatial maps. Fused per-position.
Text encoder	BioBERT (dmis-lab/biobert-base-cased-v1.2). CLS token → 256-d projection.
Tabular encoder	TabTransformer. 12 features (age, BMI, year_of_dx, days_to_followup, cigs_per_day, pack_years, alcohol, gender).
Fusion	GatedFusionTransformer (3 cross-attention layers + 2 self-attention layers). Per-modality CLS pooling. Learned modality weights.
Pathology head	5-class classifier (n_classes=5). Used by app.py + safety policy.
Staging head	4-class (no_cancer / I / II / III-IV). Anchored to v1 via KL distillation during v2 retrain — does not drift.
Risk head	Binary benign/malignant. Same KL anchor.
Segmentation decoder	SEPARATE module in <code>scripts/train_segmentation_head.py</code> . UNet-style 3-stage decoder on top of frozen ResNet50.
OOD head	Separate MLP on top of fused 256-d embedding. <code>outputs/unified_multimodal_v2/ood_head.pth</code> + <code>ood_threshold.json</code> .
Total trainable	~74.7M parameters.

Why this architecture (rationale)

ResNet50 layer4 produces a clean 7×7 spatial map — perfect for GradCAM++ and large enough to constrain via the mask-aligned attention loss. EfficientNet-B0 as the secondary backbone (rather than B4 from the original design) keeps the model deliberately smaller so val acc doesn't trivially saturate. BioBERT is the best-published clinical NLP backbone with permissive licensing. TabTransformer handles mixed categorical/continuous tabular features better than a simple MLP. The gated fusion lets the model dynamically weight modalities per-case (shown as the modality-weights radar chart in the UI).

2.2 Six-agent orchestrator

Defined in `src/agents/multimodal_orchestrator.py`. Each agent is a thin wrapper around the model exposing one capability.

#	Agent	Source file	Produces
1	UnifiedImageAgent	<code>src/agents/unified_image_agent.py</code>	ImageEvidence: pathology probs + GradCAM++ heatmap
2	TextAgent	<code>src/agents/text_agent.py</code>	TextEvidence: BioBERT attention rollout, risk_level
3	TabularRiskAgent	<code>src/agents/tabular_risk_agent.py</code>	TabularEvidence: SHAP-style perturbation importances
4	FusionReasoningAgent	<code>src/agents/fusion_reasoning_agent.py</code>	FusionDiagnosis: pathology_class, confidence, modality_weights
5	XAIAgent	<code>src/agents/xai_agent.py</code>	XAIReport: MC-Dropout uncertainty, counterfactuals
6	ClinicalRecommendationAgent	<code>src/agents/clinical_recommendation_agent.py</code>	ClinicalRecommendation: urgency, surveillance, BSG/NIC

2.3 Safety stack (added in v2)

Three independent layers, called in sequence in `app.py` around line 1370.

Layer	Module	Decision
1 — Endoscopy gate	src/app/image_atypicality.py → is_endoscopy_image()	Reject image if I-stats score < 0.55 (red dominance, tissue hue, moderate contrast)
2 — Image-atypicality detected	Same module, additional rules	Flag if heavy bleeding / deep cavitation detected even when model is confident
3a — Cross-agent consistency	src/app/cross_check.py → cross_check()	Compute coherence = geomean(class↔seg, gradcam↔seg, gradcam↔seg)
3b — Patient-safety policy	src/app/patient_safety.py → evaluate_safety()	Final gate: returns show / abstain / reject. Audit log written to outputs/audit
3c — TTA second opinion	src/app/patient_safety.py → second_opinion()	Optional strict mode requiring unanimous agreement across N augments
3d — Live-video debounces	src/app/video_pipeline.py → PolypTrack	min_persistent=3 frames, iou_thresh=0.20. No single-frame false positives

3. Data pipeline

3.1 Datasets used

Dataset	Path on disk	Purpose	Size
HyperKvasir (clean)	data/processed/hyper_kvasir_clean/	5-class pathology training	10,662 images, 23 sub-classes
CVC-ClinicDB	data/raw/CVC-ClinicDB/PNG/...	Polyp + pixel mask	612 images + masks
Kvasir-SEG	data/raw/kvasir-seg/Kvasir-SEG/	Polyp + pixel mask	1,000 images + masks
CVC-ColonDB	data/raw/test_polyp_datasets/.../CVC-ColonDB/	External polyp + mask	380 images + masks
CVC-300	data/raw/test_polyp_datasets/.../CVC-300/	External polyp + mask	60 images + masks
ETIS-LaribPolypDB	data/raw/test_polyp_datasets/.../ETIS-LaribPolypDB/	External polyp + mask	196 images + masks
Kvasir-test	data/raw/test_polyp_datasets/.../Kvasir/	External polyp + mask	100 images + masks
TCGA clinical	data/raw/tcga/clinical/clinical.tsv	Tabular features	1,802 rows × 12 features

3.2 Class label mapping

Class index	Class name (CLASS_NAMES_5)	Source sub-classes
0	polyps	polyps (HK) + CVC-ClinicDB + all polyp-mask datasets
1	uc-mild	ulcerative-colitis-grade-0-1 + grade-1 + grade-1-2 (HK)
2	uc-moderate-sev	ulcerative-colitis-grade-2 + grade-2-3 + grade-3 (HK)
3	barretts-esoph	barretts + barretts-short-segment + esophagitis-* (HK)
4	therapeutic	dyed-lifted-polyps + dyed-resection-margins (HK)

3.3 Splits + sampling

Stratified split is 75% train / 15% val / 10% test (seed=42, sklearn StratifiedShuffleSplit). Manifest CSV with MD5 hashes saved to outputs/.../manifests/split_manifest.csv on every run for audit. CVC-ClinicDB is appended to the train split AFTER MD5 dedup against HyperKvasir (to prevent CVC-ClinicDB images that exist in HK from leaking across splits). Class-balanced WeightedRandomSampler is used during v2 retraining — replacement=True, weights = 1/class_count.

3.4 Augmentation

Defined in src/data/multimodal_dataset.py: get_train_transforms(). Resize 256 → RandomCrop 224 → HFlip(0.5), VFlip(0.3), ColorJitter(b=.25,c=.25,s=.20,h=.04), RandomPerspective(.20, p=.4), ToTensor, ImageNet-normalise, RandomErasing(p=.3, scale=(.02,.10)). Tabular: Gaussian noise $\sigma=0.05$ added at train time. Polyp-mask datasets share the geometric augmentation between image and mask via shared random seed.

4. Training pipeline

Every script that produced the current checkpoints.

4.1 Training timeline (history)

When	Script	Output	Notes
v1 (Feb 2026)	experiments/run_full_pipeline.py	outputs/unified_multimodal/checkpoints/best_model.pth	Original 8 epoch train. Test acc 99.5%, but OOD is 0.86
v2 ep 1-2 (May 23)	scripts/retrain_deploy_grade.py --epochs 2	outputs/unified_multimodal_v2/checkpoints/best_model.pth	Mask-model, etc. class-balanced + KL distill from v1
v2 ep 3-4 (May 23)	Same script, warm-start from v2 --same path (overwrote)		+2 epochs. Val dipped to 0.86 but cross-val improved
Calibration	scripts/calibrate_t_ece.py	outputs/unified_multimodal_v2/temperature_calibration.pth	Grid-searched T = 0.45 for ECE-min on validation
Segmentation	scripts/train_segmentation_head.py --epochs 4	outputs/unified_multimodal_v2/seg_head.pth	U-Net decoder on frozen ResNet50. Val IoU 0.86
OOD head	scripts/train_ood_head.py	outputs/unified_multimodal_v2/ood_head.pth	High weight threshold on frozen embeddings. F1 0.86

4.2 retrain_deploy_grade.py — exact config

Optimizer	AdamW, weight_decay=0.12
LR (heads + vision)	2e-5 with CosineAnnealingLR, eta_min=2e-6
LR (BERT)	2e-6 (10× lower than rest)
Batch size	16 (MPS) / 32 (CUDA)
Loss (pathology)	FocalLoss(gamma=2.0, label_smoothing=0.1, class_weights=1/counts)
Loss (staging head)	KL(student/T teacher/T) × T ² where T=2.0
Loss (risk head)	Same KL distillation
Loss (attention)	AttentionMaskLoss: KL(softmax(features .mean(C)) downsampled_mask). Weight = 0.6 (--mask_loss_w 0.6)
Distillation weight	0.5 (--distill_w)
Warm-start	outputs/unified_multimodal/checkpoints/best_model.pth (v1)
Sampler	WeightedRandomSampler(per-class weight = 1/count, replacement=True)
Polyp+mask datasets	6: Kvasir-SEG, CVC-ClinicDB, CVC-ColonDB, CVC-300, ETIS-Larib, Kvasir-test
Train total samples	6,430 (4,082 HK+CVC + 2,348 polyp+mask)
Epochs	4 total (2 + 2 extension)
Time on MPS	~21 minutes total

4.3 Re-run training from scratch

```
# Step 1 – base v2 retrain (warm-start from v1)
python3 scripts/retrain_deploy_grade.py \
  --epochs 4 --batch_size 16 \
  --mask_loss_w 0.6 --distill_w 0.5

# Step 2 – calibrate temperature
python3 scripts/calibrate_t_ece.py

# Step 3 – train segmentation decoder
python3 scripts/train_segmentation_head.py

# Step 4 – train OOD head
```

```
python3 scripts/train_ood_head.py

# Step 5 – validate cross-vendor IoU
python3 scripts/validate_gradcam_v2.py

# Step 6 – CAde detection metrics
python3 scripts/cade_metrics_seg.py

# Step 7 – full per-class evaluation
python3 scripts/evaluate_v2_full.py
```

5. Inference pipeline

5.1 End-to-end request flow

Step	Module	What happens
1. Upload	src/app/security.validate_upload_bytes()	Size cap 10 MB, MIME allow-list, magic-byte check, PIL bomb guard.
2. Endoscopy gate	src/app/image_atypicality.is_endoscopy_image()	Pixel-stats score. Below 0.55 → reject early.
3. Model forward	UnifiedMultiModalTransformer.forward()	Image (224×224, ImageNet-normalised) + tokenised symptoms + tabular data.
4. GradCAM++	src/agents/unified_image_agent.GradCAMPlusPlus.generate()	Flows back to 50 layer4[-1]. Backward through predicted class. Up-sampled to 224×224.
5. Integrated Gradients	src/app/strong_xai.integrated_gradients()	Captum-style 24-step IG. Independent second attribution method.
6. Segmentation decoder	scripts/train_segmentation_head.SegDecoder.forward()	Optional: produces 224×224 polyp mask.
7. MC-Dropout uncertainty	src/agents/xai_agent.XAIAgent.explain()	Sets model.train(); runs 3-5 stochastic forwards; uncertainty = std.max(outputs).
8. TTA	src/app/reliability.tta_inference()	5 augmentations; agreement_pct = how many picked the modal class.
9. Cross-check	src/app/cross_check.cross_check()	Pairwise class↔seg / gradcam↔seg / gradcam↔ig. coherence = geo-mean.
10. Safety policy	src/app/patient_safety.evaluate_safety()	Returns {show, abstain, reject}. Audit-logged.
11. UI render	app.py:page_analysis() / pages/results	Verdict banner, plain-English narrative, heat-map, rationale card, chart.

5.2 Decision policy (exact thresholds)

Signal	Threshold	Source of truth
Endoscopy score	>= 0.55	SAFETY_CONFIG['min_endoscopy_score']
Confidence (calibrated)	>= 0.75	SAFETY_CONFIG['min_confidence']
MC-Dropout uncertainty	<= 0.30	SAFETY_CONFIG['max_uncertainty']
GradCAM focus	>= 0.15 (top-quartile area)	SAFETY_CONFIG['min_gradcam_focus']
Agent agreement (normal)	>= 0.66 (2 of 3)	SAFETY_CONFIG['min_agent_agreement']
Agent agreement (strict)	= 1.00 (unanimous)	SAFETY_CONFIG['strict_agent_agreement']
Cross-check coherence	>= 0.50	COHERENCE_FLOOR in cross_check.py
Live-video persistence	>= 3 frames	SAFETY_CONFIG['live_debounce_frames']
Live-video IoU	>= 0.20	SAFETY_CONFIG['live_iou_threshold']

6. Deployment architecture

6.1 Docker image (HF Space)

Defined in Dockerfile. python:3.11-slim base. Runs as uid 1000 (HF convention). Streamlit on port 7860. TRANSFORMERS_CACHE and HF_HOME both point at /home/user/.cache/huggingface. Health-check hits /_stcore/health every 30s.

6.2 HF Space configuration

Repo	Yuvraj2319/colonai (Space, type: space, SDK: docker)
Frontmatter in README.md	title, emoji, colorFrom/To, sdk=docker, app_port=7860, license=mit, short_description ≤ 60 chars
Hardware	cpu-basic (free tier)
Public	Yes (Public visibility)
Variables (visible)	COLONAI_CHECKPOINT_HF_REPO = Yuvraj2319/colonai-v2 (vestigial — checkpoint is now baked in)
Variables (visible)	COLONAI_CHECKPOINT_HF_FILE = best_model.pth
Secrets	HF_TOKEN (set programmatically via api.add_space_secret)
Storage	Checkpoint committed via Git LFS in the Space repo, NOT downloaded at runtime
Build context	Whole git tree EXCEPT what .dockerignore excludes

6.3 .dockerignore — exact whitelist pattern

```
data
experiments
paper_figures
notebooks

# Outputs: keep only the model files, exclude figures/logs/etc.
outputs/*
!outputs/unified_multimodal_v2
outputs/unified_multimodal_v2/*
!outputs/unified_multimodal_v2/checkpoints
!outputs/unified_multimodal_v2/temperature.json
outputs/unified_multimodal_v2/checkpoints/*
!outputs/unified_multimodal_v2/checkpoints/best_model.pth
```

6.4 REST API endpoints (scripts/serve_api.py)

Endpoint	Auth	Returns
GET /health	open	{status: ok, ready: True}
GET /version	open	{model_version, classes, temperature, safety_config, auth_enabled}
POST /predict	X-API-Key	{verdict, prediction, confidence, all_probs, request_id, elapsed_ms}
GET /audit/today	X-API-Key	JSONL of every prediction recorded today

7. Current metrics (everything in one place)

7.1 Cross-vendor GradCAM IoU

Dataset	Vendor	v1 IoU	v2 IoU	v1 Dice	v2 Dice
Kvasir-SEG	Olympus	0.236	0.422	0.359	0.546
ETIS-LaribPolypDB	Pentax	0.069	0.163	0.112	0.234
CVC-ColonDB	Olympus	0.109	0.205	0.181	0.301
CVC-300	Olympus	0.118	0.134	0.206	0.225
Kvasir-test	Olympus	0.213	0.415	0.325	0.546
Mean		0.149	0.268	0.237	0.371

7.2 Segmentation decoder IoU (all 5 datasets, val split)

Dataset	Vendor	IoU	Sens@IoU0.5	FPPI	mAP@0.5
Kvasir-SEG	Olympus	0.618	0.746	0.215	0.635
ETIS-LaribPolypDB	Pentax	0.609	0.383	0.444	0.359
CVC-ColonDB	Olympus	0.616	0.642	0.247	0.541
CVC-300	Olympus	0.625	0.917	0.083	0.874
Kvasir-test	Olympus	0.624	0.730	0.210	0.642

7.3 Per-class pathology metrics (val set, n=694)

Class	Precision	Recall	F1	Support
polyps	0.972	0.916	0.943	154
uc-mild	0.294	0.946	0.449	37
uc-moderate-sev	1.000	0.154	0.267	91
barretts-esoph	0.991	1.000	0.996	113
therapeutic	0.983	0.993	0.988	299

Macro-F1 = 0.729. Overall accuracy 0.865. **uc-mild over-predicted (low precision), uc-mod-sev under-predicted (low recall)**. This is the main known failure mode — see Section 8.

7.4 Calibration

Metric	Raw	After T=0.45
Expected Calibration Error (ECE)	0.168	0.062
Brier score (multi-class)	0.182	0.161

7.5 FPS benchmark (Apple Silicon MPS)

Metric	Value
--------	-------

Mean latency / frame	46.5 ms
Median latency	45.8 ms
P95 latency	48.3 ms
P99 latency	76.7 ms
Mean FPS	21.5
P95 FPS	20.7
Forward pass	42.1 ms (most of the latency)
GradCAM	0.0 ms (only runs when polyp predicted)
Live-use verdict	✓ Real-time capable (≥ 15 FPS)

8. Stage-3 problems (immediately fixable)

These can each be fixed in 1-3 sessions with the current codebase.

P-1. uc-mod-sev recall = 0.15

Diagnosis: Class-balanced WeightedRandomSampler oversamples uc-mild (only 185 samples in train) which causes the model to learn "if it looks like ulcerative colitis, call it mild." Per-class F1 = 0.267 → clinically unacceptable.

Fix: (a) Use a hierarchical loss: first classify {polyp / UC / barretts / therapeutic}, then within UC predict {mild / moderate-severe}. (b) OR: drop class-balanced sampling and use Focal loss with $\gamma=3.0$ and explicit higher class weights for uc-mod-sev. (c) OR: collect more uc-mod-sev images from public sources (HyperKvasir has 443 grade-2 + 133 grade-3 = 576; aim for ≥ 1000 by adding Kvasir-Capsule's Ulcer + Erythematous classes).

Effort: 4-8 hours retraining + evaluation

P-2. Per-polyp detection on Pentax (sens 0.38 vs 0.75 on Olympus)

Diagnosis: Segmentation IoU is at parity (0.61 both vendors), but turning a mask into a single bbox via largest-connected-component then matching to GT loses polyps that the seg decoder split into multiple components.

Fix: (a) Use a proper detection head (Faster R-CNN or YOLOv8-style anchor-free) on top of the same ResNet50 features. The seg decoder already proves the spatial features are right. (b) OR: post-process seg mask with morphological closing before bbox extraction (removes small gaps that split one polyp into two CCs). Cheap first try.

Effort: (a) 2 days + retrain · (b) 1 hour

P-3. HF Space cold-start = 30+ seconds

Diagnosis: Free CPU tier sleeps after idle; first request pays the full container boot (~2 min) including pytorch import + checkpoint load.

Fix: (a) Keep-alive: cron-job a /health hit every 10 min. (b) Upgrade to CPU-upgrade (\$0.03/h, ~\$22/month). (c) Quantize the model to int8 (cuts checkpoint size 4× and forward pass time ~2×).

Effort: (a) 15 min · (b) money · (c) 1 day quantization-aware fine-tune

P-4. Confidence calibration is on public-data val set

Diagnosis: ECE 0.062 measured on HyperKvasir val. Real hospital data has different scope brands, lighting, demographics. $T=0.45$ won't transfer perfectly.

Fix: Provide a per-site recalibration script: hospital uploads ~200 labelled samples, system re-fits T via grid search. scripts/calibrate_t_ece.py already does this — just needs a UI flow for hospital ops.

Effort: 3 hours UI + docs

P-5. OOD head trained only on synthetic OOD

Diagnosis: F1=1.0 on synthetic OOD (noise / gradients / checkerboards) but we have NO test with real OOD endoscopy images (e.g. upper-GI when expecting lower-GI, capsule endoscopy, narrow-band-imaging).

Fix: (a) Add Kvasir-Capsule images as a held-out OOD test set. (b) Add narrow-band-imaging samples from CVC-ClinicNBI. (c) Re-train OOD head with these as additional OOD class — F1 will drop honestly but the score will be meaningful.

Effort: 6 hours data prep + retraining

9. Stage-4 problems (deeper research)

These need real research, not just engineering. Months, not days.

R-1. No real cancer staging data

Diagnosis: The staging head outputs {no_cancer / I / II / III-IV} but the entire training corpus is screening-stage (HyperKvasir, CVC). TCGA gives us tabular features but no images. The staging head learns to output "no_cancer" always, and we anchor it via KL distillation. So the staging head is essentially a vestigial output, kept only because it was in v1's architecture.

Research direction: (i) Acquire a histopathology image dataset (PAIP, PANDA, TCGA-COAD whole-slide images). (ii) Train staging head on histology + transfer to endoscopy via feature alignment. (iii) OR drop the staging head and present a more honest two-output system: pathology class + binary cancer-risk.

R-2. Per-site distribution shift not measurable

Diagnosis: Vendor bias was the v1 problem; v2 closed it. But there's a deeper operator-induced bias: different endoscopists use different insertion technique, different bowel-prep quality, different photographic style. We can't measure this without per-site labelled data.

Research direction: Federated-learning pilot: 3-5 hospitals each upload class probabilities (not images) for their own cases over 30 days; we measure prediction-shift across sites. Use that to motivate per-site fine-tuning vs. global model.

R-3. No characterisation (CADx), only detection (CAdE)

Diagnosis: We say "polyp" but not "adenoma vs hyperplastic vs serrated". A clinician still needs to perform biopsy because we don't differentiate histology types. CADx (computer-aided diagnosis) is the actual clinical workflow target.

Research direction: (i) Get NICE polyp classification training data (Kudo Pit Pattern, NBI International Colorectal Endoscopic classification). (ii) Add a polyp-type head with 3-4 classes. (iii) Validate against histology gold standard. This is a separate model+study, probably worth its own dissertation chapter.

R-4. Live-video temporal model is just frame-by-frame + tracker

Diagnosis: Each frame is independently classified. A proper video model would use temporal convolutions (X3D, TimeSformer) to exploit motion cues — e.g. how a polyp moves differently from a fold when the scope moves.

Research direction: Train a 16-frame temporal classifier on Kvasir-Capsule sequences + SUN-SEG. Compare against per-frame baseline on both detection F1 and temporal consistency.

R-5. No active-learning loop with hospital feedback

Diagnosis: Once deployed, the system doesn't learn from doctor corrections. Every "abstain" case that a doctor labels is wasted training signal.

Research direction: Add a 'doctor feedback' button in the UI (yes/no/which class). Store the feedback + image SHA-256 in an append-only log. Monthly auto-retrain on accumulated corrections. Needs careful PHI handling — image hashes only, no patient identifiers.

R-6. No proper safety validation in a clinical setting

Diagnosis: We measured cross-vendor IoU on PUBLIC datasets. We have not measured what happens when a real endoscopist uses the system on real patients over 100 real procedures. That's the only number that matters for clinical

adoption.

Research direction: Partner with a hospital for a 3-month observational pilot. Measure (i) operator-perceived agreement, (ii) overlooked-polyp rate with vs without ColonAI, (iii) confidence-calibration on hospital data, (iv) operator time-to-decision. Standard PRISMA-AI reporting.

10. Improvements roadmap (prioritised)

Priority	Item	Effort	Impact
P0 — do first	Fix uc-mod-sev recall (Section 8 P-1)	1 week	Removes the single worst clinical failure.
P0	Per-site recalibration UI flow (P-4)	3 hours	Required before any real hospital deploy.
P0	Morphological closing before seg→bbox (P-2 part b)	1 hour	Free Pentax detection boost.
P1	Proper detection head (Faster R-CNN/YOLOv8) (P-2 a)	2 days	Brings Pentax detection F1 to ~0.6.
P1	Real OOD test set (Capsule + NBI) (P-5)	6 hours	Makes the OOD head's F1 meaningful.
P1	Doctor-feedback button + log (R-5)	1 day	Enables active learning loop.
P2	int8 quantisation for HF Space cold-start (P-3 c)	1 day	Cuts checkpoint to 150 MB, doubles FPS.
P2	Temporal model for live video (R-4)	2 weeks	Improves live-feed detection.
P2	CADx polyp typing head (R-3)	1 month	Closes the gap to clinical workflow.
P3	Federated cross-site distribution audit (R-2)	3 months	Needed for multi-site deployment.
P3	Drop or replace staging head (R-1)	1 day	Honesty win; minor functional change.
P3	Clinical observational pilot (R-6)	3 months	Required for any regulatory path.

Suggested order for a 6-month follow-up dissertation

(1) Month 1: P0 items + P1 detection head — measurable lift in headline numbers. (2) Month 2-3: CADx polyp typing (R-3) — new contribution. (3) Month 4-5: Real-hospital pilot (R-6) — only honest clinical validation. (4) Month 6: Write everything up, including the negative results.

11. Operational runbook

11.1 Restart the live Space

```
# From your local venv (huggingface_hub installed):
python3 -c "from huggingface_hub import HfApi; HfApi(token=open('~/.cache/huggingface/token').read().strip()).restart_space(repo_id='Yuvraj2319/colonai')"

# OR via curl:
curl -X POST -H "Authorization: Bearer $(cat ~/.cache/huggingface/token)" \
https://huggingface.co/api/spaces/Yuvraj2319/colonai/restart
```

11.2 Push a code change to the live Space

```
# GitHub push is the source of truth:
git push origin main

# HF Space repo is a separate git remote – push there too:
git remote add space https://huggingface.co/spaces/Yuvraj2319/colonai # one time
git push --force space main # use the token from ~/.cache/huggingface/token as password
```

11.3 Re-upload the checkpoint to the Space

```
# The checkpoint lives in the Space repo via Git LFS.
# To replace it after retraining:

git clone https://huggingface.co/spaces/Yuvraj2319/colonai /tmp/space
cd /tmp/space && git lfs install
cp ~/Desktop/.../outputs/unified_multimodal_v2/checkpoints/best_model.pth \
  outputs/unified_multimodal_v2/checkpoints/best_model.pth
git add outputs/unified_multimodal_v2/checkpoints/best_model.pth
git commit -m "new checkpoint"
git push
```

11.4 Debug a Space that won't load the model

1. Open <https://huggingface.co/spaces/Yuvraj2319/colonai/logs> and select Container.
2. Look for the [STARTUP] lines (first 10 lines of output):
[STARTUP] COLONAI_CHECKPOINT_HF_REPO = ...
[STARTUP] HF_TOKEN present = ...
3. Then look for [CHECKPOINT] lines from `_maybe_download_checkpoint()`.
4. If "COLONAI_CHECKPOINT_HF_REPO is unset → demo mode" → env vars not set.
5. If "v2 checkpoint already present" → file is baked into image. Verify size.
6. If "HF Hub download failed" → check token, check model repo public.

If env vars are missing, set them via:

```
from huggingface_hub import HfApi
api = HfApi(token=TOKEN)
api.add_space_variable(repo_id='Yuvraj2319/colonai',
  key='COLONAI_CHECKPOINT_HF_REPO', value='Yuvraj2319/colonai-v2')
```

11.5 Run the local Streamlit app for development

```
cd ~/Desktop/Agentic_Multimodal_Colon_Cancer_AI\ copy
source venv/bin/activate
streamlit run app.py --server.port 8502 --server.address 127.0.0.1
```

11.6 Run the REST API locally

```
# Without auth (dev):
python3 scripts/serve_api.py

# With auth (any deploy):
export COLONAI_API_KEY=$(openssl rand -hex 32)
python3 scripts/serve_api.py
```



```
# Test:
curl -X POST -H "X-API-Key: $COLONAI_API_KEY" \
  -F image=@some_polyp.jpg http://127.0.0.1:8081/predict
```

12. Code navigation index

Where to look when you want to change X.

To change...	Edit...
Safety thresholds (confidence/uncertainty/etc)	src/app/patient_safety.py — SAFETY_CONFIG dict (top of file)
Cross-check thresholds	src/app/cross_check.py — COHERENCE_FLOOR, SEG_MIN_POLYP_AREA const
Endoscopy gate threshold	src/app/image_atypicality.py — is_endoscopy_image() default threshold=0.55
Live-video debouncer parameters	src/app/video_pipeline.py — PolypTracker(...) constructor calls (lines 335, 472)
Upload size cap	src/app/security.py — MAX_UPLOAD_BYTES (default 10 MB) + .streamlit/config.toml
Training hyperparameters	scripts/retrain_deploy_grade.py — argparse defaults in main()
Class label map (5 classes)	src/data/multimodal_dataset.py — SUBCLASS_TO_LABEL dict
Plain-English narrative templates	src/app/patient_ui.py — PLAIN_NAMES + NEXT_STEPS_PLAIN dicts
Result-page rendering	app.py — page_analysis() function around line 3100
Sidebar navigation + status	app.py — render_sidebar() function around line 1700
Add a new agent	src/agents/ + register in src/agents/multimodal_orchestrator.py __init__
Add a new chart to results	app.py — page_analysis() in the section that creates Plotly figures
Change the safety verdict banner copy	src/app/patient_ui.py — verdict_card_html()
Add a new dataset for training	scripts/retrain_deploy_grade.py — POLYP_MASK_DATASETS list (top)
Change Docker base image / port	Dockerfile (root of repo)
Change what's deployed to HF Space	Edit on local, push to both origin (GitHub) AND space (HF Space) remotes
Add/remove env vars on the Space	huggingface_hub.HfApi.add_space_variable / add_space_secret (Python)

13. Lessons learned (deploy gotchas)

Things that wasted hours during this build — do not repeat.

- **HF Spaces removed Streamlit from the create-Space UI dropdown.**

Pick Docker SDK and write a Dockerfile that runs Streamlit. Don't waste time looking for the Streamlit option — it's gone from the UI, even though it still works via README frontmatter `sdk: streamlit`.

- **HF Spaces' `upload_file()` / `create_commit()` silently dedupes large blobs via XET.**

If a file's SHA-256 already exists in another HF repo you own, `upload_file` may return success with a commit URL but the content blob never actually lands. Use git CLI with `git-lfs` installed instead, OR modify a single byte to break the dedup.

- **`.gitignore` on the Space repo can SILENTLY refuse git add.**

When I tried to add the checkpoint via git CLI, `.gitignore` had `'outputs/'` which blocked the add — but the `upload_file` path didn't error. Always run ``git status --short`` and ``git lfs ls-files`` to confirm the file is staged before celebrating a successful push.

- **`torch.load` default is `weights_only=False` — pickle RCE if checkpoint is untrusted.**

Use `src/app/security.safe_torch_load()` which prefers `weights_only=True` and falls back to `allow_unsafe=True` only for first-party files.

- **Streamlit ignores `.streamlit/config.toml`'s ``address`` when you pass `--server.address`.**

Don't trust the config file alone; always set `--server.address 127.0.0.1` explicitly.

- **`PIL.Image.open()` has no built-in size limit by default.**

A 30 KB PNG can legally decode to 1 GB in RAM (decompression bomb). Import `src.app.security` at the top of any module that calls `Image.open()` — it sets `MAX_IMAGE_PIXELS = 100M` as a side effect.

- **Streamlit's ``unsafe_allow_html=True`` is XSS-vulnerable for user-typed strings.**

Always pass user input through `src.app.security.escape_html()` before interpolating into HTML. `patient.get('name')` was leaking unescaped into the narrative — fixed, but watch for new callsites.

- **HF Hub model download counter has a multi-hour delay.**

Don't use `download_count` to verify whether the Space is fetching your checkpoint. Add `[STARTUP]` / `[CHECKPOINT]` log prints instead.

- **Streamlit caching prevents module-level code from re-running on script rerun.**

MOSTLY false — module-level code DOES re-run on script rerun in Streamlit. But `@st.cache_resource`'d functions run once per server lifetime. If you change model-loading code, you must restart the Streamlit server.

- **EMA optimizer crashes on BERT-layer unfreeze.**

`EMA.update()` must check ``if n not in self.shadow: init``. This was a v1 bug — the fix is committed in `experiments/run_full_pipeline.py`.

14. Reference: env vars, paths, URLs, hashes

14.1 Environment variables

Variable	Default	Where read	Purpose
COLONAI_CHECKPOINT_HF_REPO	(unset)	app.py:67	HF Hub model repo to download best_model.pth from. Verifies file exists.
COLONAI_CHECKPOINT_HF_FILENAME	best_model.pth	app.py:68	Filename inside the model repo to download.
HF_TOKEN	(unset)	app.py:69, hf_hub_download.py	Authenticate HF Hub access. Silences rate-limit warnings.
HUGGINGFACE_TOKEN	(unset)	app.py:69 (fallback)	Alternative name for HF_TOKEN.
COLONAI_API_KEY	(unset)	scripts/serve_api.py + src/app.py:101	App security key and the API to 0.0.0.0. Enforced on /predict + /audit.
COLONAI_BIND	127.0.0.1	scripts/serve_api.py	API bind host. Refuses 0.0.0.0 unless COLONAI_API_KEY is set.
COLONAI_PORT	8081	scripts/serve_api.py	API port.
COLONAI_CORS_ORIGINS	localhost:8501,localhost:8500	scripts/serve_api.py	Comma-separated CORS allow-list.
COLONAI_EXPOSE_DOCS	(unset)	scripts/serve_api.py	When set, enables Swagger /docs.
COLONAI_LOG_LEVEL	INFO	scripts/serve_api.py	Log verbosity.
TRANSFORMERS_CACHE	/home/user/.cache/huggingface	Default	Where HF Hub caches transformer models.
HF_HOME	/home/user/.cache/huggingface	Default	HF Hub root cache.

14.2 Key file paths

Path	Purpose
outputs/unified_multimodal_v2/checkpoints/best_model.pth	v2 active checkpoint (572 MB)
outputs/unified_multimodal_v2/seg_head.pth	UNet segmentation decoder
outputs/unified_multimodal_v2/ood_head.pth	OOD detector MLP
outputs/unified_multimodal_v2/ood_threshold.json	OOD decision threshold (F1-optimal)
outputs/unified_multimodal_v2/temperature.json	Calibration temperature T = 0.45
outputs/unified_multimodal_v2/metrics_v2.json	Per-class P/R/F1, confusion matrix, ECE
outputs/unified_multimodal_v2/cross_vendor_gradcam_compare.json	v1 vs v2 GradCAM IoU per dataset
outputs/unified_multimodal_v2/seg_iou.json	Segmentation IoU per epoch per dataset
outputs/unified_multimodal_v2/cade_metrics_seg.json	CADe sensitivity / FPPI / mAP per dataset
outputs/unified_multimodal_v2/fps_benchmark.json	FPS measurement output
outputs/unified_multimodal_v2/figures/	Dissertation plots + overlays
outputs/unified_multimodal_v2/agent_coherence_report.json	3-case deployment-readiness test report
outputs/audit/audit_YYYYMMDD.jsonl	Daily audit log — every prediction (chmod 0600)
data/processed/hyper_kvasir_clean/	HyperKvasir training data (lower/upper-gi-tract subfolders)
data/raw/CVC-ClinicDB/PNG/{Original,Ground Truth}/	CVC-ClinicDB images + masks
data/raw/test_polyp_datasets/TestDataset/	External polyp datasets (Pentax + others)

data/raw/tcga/clinical/clinical.tsv

TCGA tabular features

14.3 URLs and remotes

Source code (GitHub)	https://github.com/Yuvraj235/Agentic_Multimodal_Colon_Cancer_AI
Live demo (HF Space)	https://huggingface.co/spaces/Yuvraj2319/colonai
Model checkpoint (HF Hub)	https://huggingface.co/Yuvraj2319/colonai-v2
Direct Streamlit URL	https://yuvraj2319-colonai.hf.space
HF tokens settings	https://huggingface.co/settings/tokens

14.4 Recent commit history (main branch)

SHA	Message
cb797a2	docs: buyer-ready ColonAI_Pitch.pdf
063add7	fix(spaces): include outputs/unified_multimodal_v2/* in Docker image
03c4e6b	diagnostic: print env var visibility on app startup
fe0bfc0	fix(spaces): surface checkpoint download status in the sidebar
d2ab9fc	fix(spaces): make HF Hub checkpoint download LOUD + token-aware
6e76344	docs: add live demo link to README — Space is live on HF
a5edbec	fix(spaces): shorten short_description to ≤ 60 chars
df33146	deploy: Dockerfile for Hugging Face Spaces
519db5c	deploy: auto-download checkpoint from HF Hub + clearer Spaces setup
8536cc3	docs: patient-facing README + HF Spaces deploy config
b4f3b56	Security hardening + UI safety + cross-check consistency net
db103e0	v2 deployment polish: segmentation head, OOD, REST API, safety, metrics
20f66cc	v2 retrain: mask-aware, multi-vendor, patient-safety-grade
b1554c3	Update GradCAM++ section: use sample_003_polyps image as fig06
581e0fa	Update GradCAM++ figure captions with accurate image-verified descriptions

14.5 HF Hub auth

Token stored at `~/.cache/huggingface/token` (write scope). Get a new one at <https://huggingface.co/settings/tokens>. Run `huggingface-cli login` to refresh. **DO NOT** commit this token to git or paste it into chat logs.

This document is the source of truth for the technical state of ColonAI. If you change a threshold, retrain the model, or alter the deployment topology, regenerate this PDF (`python3 scripts/make_handover_pdf.py`) and replace the version in the repo.

Document author: Yuvraj Pratap Singh (Amity University). Built with extensive AI assistance — every line in this codebase has been human-reviewed.